

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper: CS-105 — Discrete Mathematics

Time: Three hours

Maximum Marks: 70

Instructions: ATTEMPT ANY FIVE QUESTIONS. THE FIGURES IN THE RIGHT-HAND MARGIN INDICATE

Question 1.

- (a) Define a set. Give another representation of the following sets: [4]
- (i) $\{x \mid x \text{ is an integer and } 5 \leq x \leq 12\}$
 - (ii) $\{2, 4, 8, \dots\}$
 - (iii) All the countries of the world.
 - (iv) $\{x \mid x \text{ is an integer and } x^2 = 2\}$
- (b) Define number theoretic, total, and partial function. Give one example of each. [6]
- (c) Let a relation be defined over a finite set. Answer the following considering the matrix representation of the relation: [4]
- (i) How to check that the relation is a function.
 - (ii) How to check that the relation is a one-to-one function.
 - (iii) How to check that the relation is an onto function.
 - (iv) How to check that the relation is a one-to-one and onto function.

Question 2.

- (a) Define partial order relation, POSET, and Lattice. [4]
- (b) Let $X = \{1, 2, 3, 4, 6, 8, 12, 24\}$ and the relation $\leq$ be such that $x \leq y$ if x divides y . Answer the following questions for the POSET $(X, \leq)$: [10]
- (i) List all ordered pairs of relation $\leq$.
 - (ii) Give relation matrix of the relation $\leq$.
 - (iii) Find the matrix of inverse relation of $\leq$ from matrix of relation $\leq$.
 - (iv) Draw the Hasse diagram of $(X, \leq)$.
 - (v) Find the maximal and minimal element, if it exists.
 - (vi) Find the greatest and least element, if they exist.
 - (vii) Find all upper bounds of $\{2, 6\}$.

- (viii) Find the least upper bound of $\{2, 6\}$, if it exists.
- (ix) Find all lower bounds of $\{4, 6\}$.
- (x) Find the greatest lower bound of $\{4, 6\}$, if it exists.

Question 3.

- (a) Show that the conclusion “If I do not finish writing the program, then I will wake up feeling refreshed”, logically follows from the premises: “If you send E-mail message, then I will finish writing the program,” “If you do not send me an E-mail message, then I will go to sleep early,” and “If I will go to sleep early, then I will wake up feeling refreshed.”
[4]
- (b) Write the negation of the premises and conclusion given in 3(a). [4]
- (c) Describe the procedure proof by contradiction. Using the procedure proof by contradiction, show that the conclusion Q logically follows from the premises P and $P \rightarrow Q$.
[6]

Question 4.

- (a) Indicate the variables that are free and bound in the following statements. Also show the scope of the quantifiers: [5]
 - (i) $(\forall x)P(x, y)$
 - (ii) $(\forall x)(P(x) \rightarrow Q(x))$
 - (iii) $(\forall x)(P(x) \rightarrow (\exists y)R(x, y))$
 - (iv) $(\forall x)(P(x) \rightarrow R(x)) \vee (\forall x)(P(x) \rightarrow Q(x))$
 - (v) $(\exists x)P(x) \wedge Q(x)$
- (b) Discuss the limitation of propositional calculus, and how it is overcome in predicate calculus. [4]
- (c) Suppose the domain of the propositional function $P(x)$ consists of the finite set $\{x_1, x_2, x_3, \dots, x_n\}$. Express the following statements without using quantifiers: [5]

$$\exists x P(x), \quad \forall x P(x), \quad \neg \exists x P(x), \quad \exists x \neg P(x), \quad \forall x \neg P(x)$$

Question 5.

- (a) Define an Euler circuit in a graph. Give necessary and sufficient condition that a connected graph have an Euler circuit. [3]
- (b) Does the Hasse diagram given in question 2(b)(iv) have an Euler circuit? If yes, give at least one Euler circuit present in this graph. [2]
- (c) Define Hamiltonian circuit in a graph. If they exist, give necessary and sufficient conditions (or necessary or sufficient condition) that a connected graph have a Hamiltonian circuit. [3]
- (d) Prove that in a graph the number of nodes with odd degree must be even. [6]

Question 6.

- (a) Can five houses be connected to two utilities without connection crossings? Justify your answer with proper reasoning. [4]
- (b) For which values of n the following graphs are planar: [4]
- (i) K_n
 - (ii) W_n
 - (iii) $K_{m,n}$
 - (iv) Q_n
- (c) For which value of n are these graphs bipartite? And also find their chromatic number: [4]
- (i) K_n
 - (ii) C_n
 - (iii) $K_{m,n}$
 - (iv) Q_n
- (d) Find the chromatic number of the Hasse diagram of the POSET given in question 2(a). [2]

Question 7.

- (a) Define phrase-structured grammar and describe the Chomsky classification of phrase-structured grammar. [5]
- (b) Let $G = (V, T, P, S)$ be the phrase-structured grammar with $V = \{0, 1, A, B, S\}$, $T = \{0, 1\}$, and set of productions $P = \{S \rightarrow 0A, S \rightarrow 1A, A \rightarrow 0B, B \rightarrow 1A, B \rightarrow 1\}$. Answer the following: [9]
- (i) Show that 10101 belongs to the language generated by G .
 - (ii) Show that 10110 does not belong to the language generated by G .
 - (iii) What is the language generated by G ?

Question 8.

- (a) Give formal definition of a finite state machine which produces an output. Model a two-input logical AND gate using any finite state machine. [6]
- (b) Show that the total number of one-to-one functions from A to B is $n(n-1)(n-2) \dots (n-m+1)$, where $|A| = m$ and $|B| = n$. [3]
- (c) Model any problem of your choice using a recurrence relation and also give the solution of the modelled problem. [5]

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